Conor Wood Hayes

cwoodhayes.github.io · github.com/cwoodhayes · linkedin.com/in/cwoodhayes · cwhayes@u.northwestern.edu · (949) 554 3392

Education

Northwestern University – *M.S. Robotics* — Sep 2025–Aug 2026 (expected)

University of Southern California – *B.S. Computer Engineering & Computer Science* — 2015–2019
Magna Cum Laude. Minor in Chinese for the Professions, Thematic Option (Honors in Liberal Arts)

Skills

Programming Languages: Python, C++, C, CMake, Bash/Zsh, Rust, SQL, TypeScript, Verilog, Assembly
Robotics: SLAM, Computer Vision, Tactile Sensing, State Estimation, Robot Kinematics & Dynamics, Path Planning, Control Theory
Machine Learning & AI: Imitation Learning, Foundation Models (LLM/VLM/VLA), Deep Learning, Deep Reinforcement Learning
Software: ROS 2, PyTorch, NumPy, Pandas, Linux, Git, Docker, AWS, CI/CD, Databases
Hardware: PCB Design, Electronics, Microcontrollers, Motors, CAD (OnShape), 3D Printing, Laser Cutting

Work Experience

Conor Hayes Software Consulting, Inc | *CEO* — 2023 - 2025, IN/NY/CA
- Built reusable hardware testing framework (Python) for two $500K nuclear fusion test racks, driving on-time delivery in 8mos.
- Redesigned BLE protocol (C, TypeScript) for NRF52-based wearable. +2 sensors, +800% bandwidth, -10% power consumption.
- Prototyped a low-latency peer-to-peer telepresence platform for controlling robots via web client (WebRTC, AWS, TypeScript).

Wesper, Inc | *Software Engineer (8th employee @ startup)* — 2020 - 2022, New York City
- Developed sleep apnea diagnosis wearable from prototype to FDA cleared mass product as lead firmware developer (C, NRF5X).
- Added new sensors, on-device algorithms, and BLE services to enable real-time reporting of heart rate, SpO2, and sleep pose.
- Led a team of 3 to build backend from scratch (AWS ECS, EC2, λ, S3, RDS MySQL, Python) to ingest and process all patient data.

Honeybee Robotics | *Software & Electrical Engineering Intern* — May - Aug 2018, Pasadena, CA
- Developed drivers, middleware (C++, ZeroMQ), and GUIs (Qt), for ROS-based testing framework for space exploration robotics.
- Designed, assembled, and validated PCB (Altium) for DC motor control of weatherproof hand drill for arctic exploration.

NASA Jet Propulsion Laboratory | *Flight Electronics Intern (Group 349E) - Sphinx Project* — Jan - Aug 2017, Pasadena, CA
- Developed still-in-use automated test system & BSP modules (C, Python, assembly) for first deep-space cubesat avionics system.
- Supported initial board bring-up, discovered critical bugs in novel rad-hardened NAND flash controller & other peripherals.

Projects

PolyUMI (CR2 @ ICRA 2026)— Visual-Tactile Data Collector for Dextrous Manipulation Imitation Learning — Jan-Aug 2026
- Designed & built wireless handheld grasping data collector (based on UMI) with novel sensor array & custom optical tactile sensor.
- Developed training, inference, and visualization pipeline (Python, ROS 2) for multimodal diffusion policy on 7DoF arm.

HANDyDriver v3 — High-Performance Brushless DC Motor Controller for Dextrous Hands — Apr-Aug 2026
- Designed & validated 3rd-gen PCB for 24V BLDC motor control (KiCAD), shrinking board size and adding support for EtherCAT.
- Ported field-oriented control (FOC) library from Arduino to dual-core NXP MCU, for simultaneous 10kHz control of 3 motors (C++).

LeHome Challenge 2026 — Vision-Language-Action (VLA) Model for Bimanual Garment Folding — Jan-Apr 2026
- Led team of 7 to train 2 SO-101 arms to fold laundry, placing in top 20% of 250 teams in global competition with 40% task success.
- Trained and evaluated various models (SmolVLA, X-VLA, Lingbot, DP, ACT) with augmented teleop data in Isaac Sim environment.

Simultaneous Localization and Mapping (SLAM) from Scratch — Jan-Mar 2026
- Designed, developed, & deployed from-scratch Extended Kalman Filter (EKF) LiDAR SLAM pipeline for Turtlebot3 (ROS2, C++).
- Implemented custom RViz-based simulation environment to test kinematics, control, data-association, and SLAM algorithms.

PenPal — Handwriting with the Franka Panda 7DoF Robot Arm — Nov-Dec 2025
- Designed system architecture, led group of 4 to create a robot arm-based handwriting system for a randomly moving whiteboard.
- Developed online, reactive trajectory generation and closed-loop visual cartesian control (Python, ROS 2, MoveIt 2, RealSense).

Traveler IV — First 100% undergrad-made rocket to fly to space *(USC Rocket Propulsion Lab)* — 2015-2019
- Built & led 30-person avionics team overseeing all software & electronics to manufacture and fly high-performance rockets.
- Developed flight software (C++), PCB's (Altium), EGSE (Python, ham radio), and data analysis (Matlab) for custom avionics system.

Awards + Hobbies

Awards: AIAA Achievement Award (2019), NAE/USC Grand Challenges Scholar (2019), USC Rennaisance Scholar (2019)
Music: Accumulated 800,000+ Spotify streams on original music under artist name *Wise John* (24 songs, 1 album, 1 EP).
Languages: English (native), Mandarin Chinese (conversational)